Monopole triangle over integers

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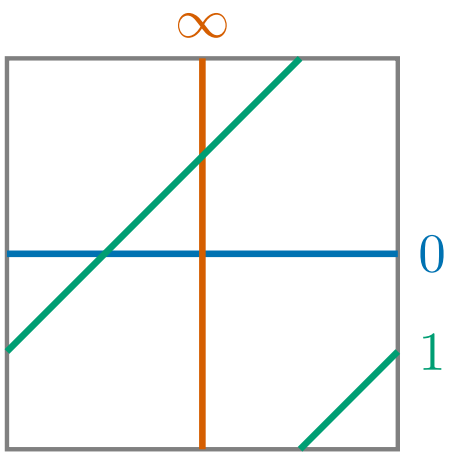
Surgery triad

Definition

Let M be a compact oriented 3-manifold with $\partial M \cong T^2$. Choose oriented curves $\gamma_i \subset \partial M$, $i \in \mathbb{Z}/3$, with

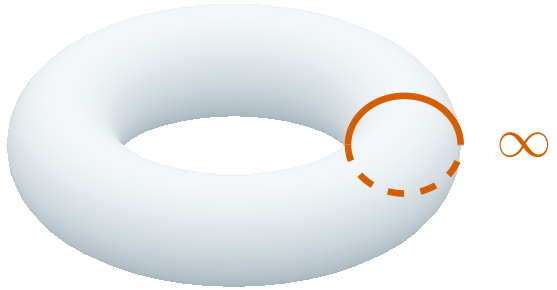
$$\gamma_0 \cdot \gamma_1 = \gamma_1 \cdot \gamma_2 = \gamma_2 \cdot \gamma_0 = -1.$$

The fillings $Y_i = M(\gamma_i)$ form a **surgery triad**. Let $W_i : Y_i \to Y_{i+1}$ be the corresponding 2-handle cobordism.



∂M

gluing



$V = D^2 \times S^1$

Surgery exact triangle

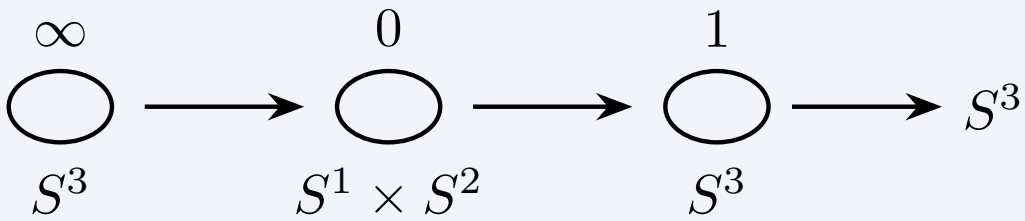
Definition

Let $F : \mathrm{Cob} \to \mathrm{grMod}$ be a functor, sending closed oriented connected 3-manifolds to graded modules and cobordisms to module morphisms. It satisfies the **surgery exact triangle property** if every surgery triad gives an exact triangle

$$F(Y_0) \xrightarrow{F(W_0)} F(Y_1) \xrightarrow{F(W_1)} F(Y_2) \xrightarrow{F(W_2)} F(Y_0).$$

That is, for $i \in \mathbb{Z}/3$, $\operatorname{im} F(W_i) = \ker F(W_{i+1})$.

Example (Unknot)



In Heegaard Floer homology over $\mathbb{F}_2 = \mathbb{Z}/2$,

$$\mathbb{F}_2 \xrightarrow{\binom{1}{0}} \mathbb{F}_2^2 \xrightarrow{(0,1)} \mathbb{F}_2 \xrightarrow{0} \mathbb{F}_2.$$

Surgery generation

Theorem (Baldwin–Bloom, Culler '14, Ni '25)

Using the rule that two members of a surgery triad generate the third, every closed oriented connected 3-manifold is generated from S^3.

Corollary (used by Lin '17)

Let $F, G : \mathrm{Cob} \to \mathrm{grMod}$ be functors satisfying the surgery exact triangle property. If a natural transformation $\Psi : F \Rightarrow G$ has $\Psi(S^3)$ an isomorphism, then $\Psi(Y)$ is an isomorphism for every closed oriented connected 3-manifold Y.

Proof. Five lemma and induction.

$$\begin{array}{ccccccccc} F(Y_0) & \longrightarrow & F(Y_1) & \longrightarrow & F(Y_2) & \longrightarrow & F(Y_0) & \longrightarrow & F(Y_1) \\ \cong\downarrow & & \cong\downarrow & & \downarrow{\scriptstyle \Psi(Y_2)} & & \cong\downarrow & & \cong\downarrow \\ G(Y_0) & \longrightarrow & G(Y_1) & \longrightarrow & G(Y_2) & \longrightarrow & G(Y_0) & \longrightarrow & G(Y_1) \end{array}$$

Montesinos trick

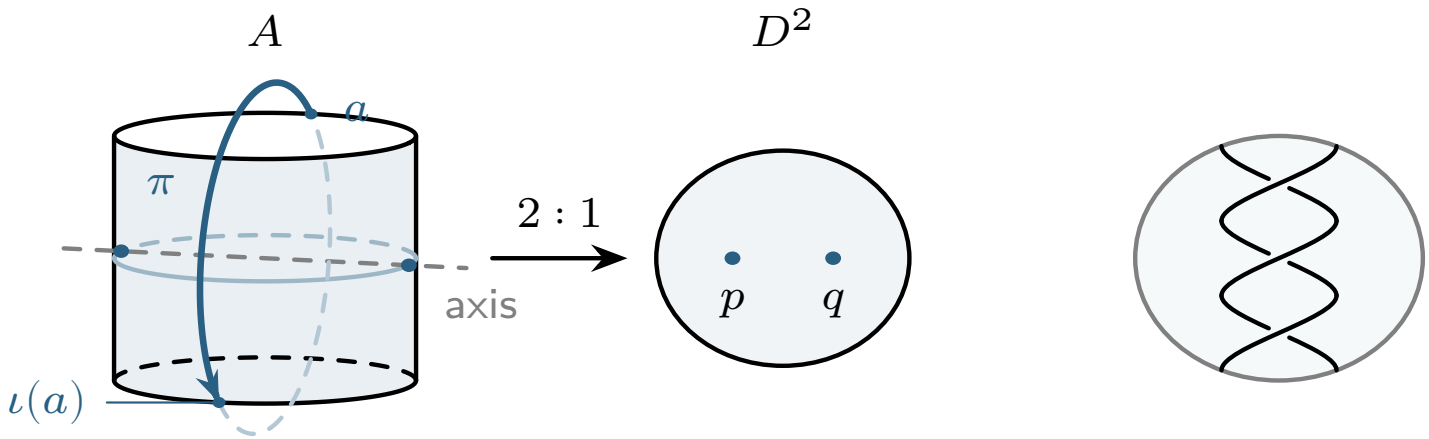
Definition

The annulus $A = S^1 \times [-1, 1]$ has involution

$$\iota(e^{i\theta}, t) = (e^{-i\theta}, -t).$$

Its quotient is D^2, with two branch points. Thus $A \times I \cong S^1 \times D^2 \to D^2 \times I$ is branched over two trivial arcs.

A **rational tangle** is a pair (B^3, τ) homeomorphic to $(D^2 \times I, \{p, q\} \times I)$, where $p, q \in \operatorname{int} D^2$ are distinct. Every rational tangle arises from this quotient by choosing a boundary parametrization.



two fixed points | branch set $\{p, q\}$ | 3 half-twists

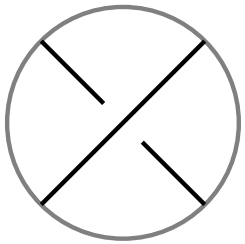
Montesinos trick

Proposition (Montesinos)

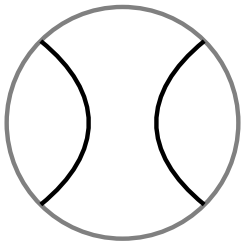
Rational tangle replacement lifts to Dehn filling. For the displayed crossing of L with the two **resolutions** L_0, L_1,

$$(-\Sigma_2(L), -\Sigma_2(L_0), -\Sigma_2(L_1))$$

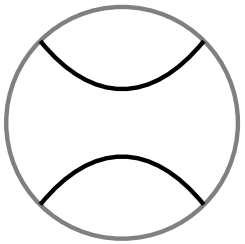
is a surgery triad.



L



L_0



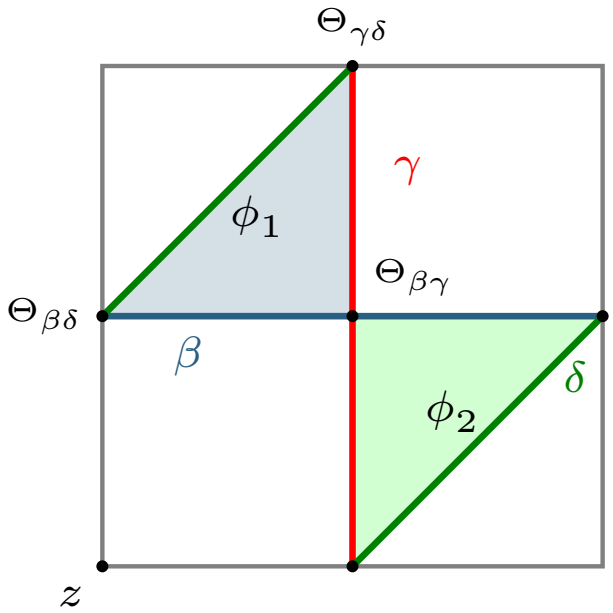
L_1

Floer exact triangles

Theory	**Coefficients**	**Authors**
Instanton	$\mathbb{Z}$	Floer '90, with extra bundle data on Y_2, W_1, W_2
Heegaard Floer	$\mathbb{F}_2, \mathbb{Z}$	Ozsváth–Szabó '04 $(\widehat{HF}, HF^+)$
Completed HF^-	$\mathbb{F}_2[[U]]$	Manolescu–Ozsváth '10
Monopole	$\mathbb{F}_2$	Kronheimer–Mrowka–Ozsváth–Szabó '07
Monopole	$\mathbb{Z}[\sqrt{-1}]$	Freeman '21, with local coefficients
Monopole	$\mathbb{Q}$	Lin–Ruberman–Saveliev '23
Heegaard Floer	$\mathbb{Z}$	Abouzaid–Manolescu '25, with two twisted cobordism maps
Monopole spectra $(b_1(Y_i) = 0)$		Sasahira–Stoffregen (in preparation), mentioned in Dai–Sasahira–Stoffregen '26

- Ozsváth–Szabó and Lin–Ruberman–Saveliev use auxiliary signs.
- Freeman and Abouzaid–Manolescu use local coefficients.

Necessity of local coefficients



The two triangles have the same sign (Abouzaid–Manolescu '25).

$$\epsilon(\phi_1) + \epsilon(\phi_2) = \pm 2.$$

In $\widehat{HF}$,

$$F_{\beta\gamma\delta}(\Theta_{\beta\gamma} \otimes \Theta_{\gamma\delta}) = \pm 2\Theta_{\beta\delta}.$$

By the A_∞ associativity relation,

$$F(W_1) \circ F(W_0) \simeq \pm 2F_{\alpha\beta\delta}(- \otimes \Theta_{\beta\delta}).$$

The same happens in bordered Floer theory over $\mathbb{Z}$ (Knowles–Petkova '21).

Obstruction in monopole theory

The cocore of W_0 and the core of W_1 form the exceptional sphere

$$E = D_0^{cc} \cup D_1^{c},\ E \cdot E = -1$$

in a $\overline{\mathbb{CP}}^2$ summand of $W_0 \cup_{Y_1} W_1$.

Complex conjugation on this summand extends by the identity. It preserves the orientation and homology orientation and sends $[E]$ to $-[E]$.

Fix the Spin^c structure on the complement. Conjugation sends $\mathfrak{s}_m$ to $\mathfrak{s}_{-m}$.

$$\langle c_1(\mathfrak{s}_{\pm m}), [E] \rangle = \pm m,\ m \in 2\mathbb{Z} + 1.$$

The untwisted contributions have the same sign. Using local coefficients over $\mathbb{Z}[u, u^{-1}]$, the weights are

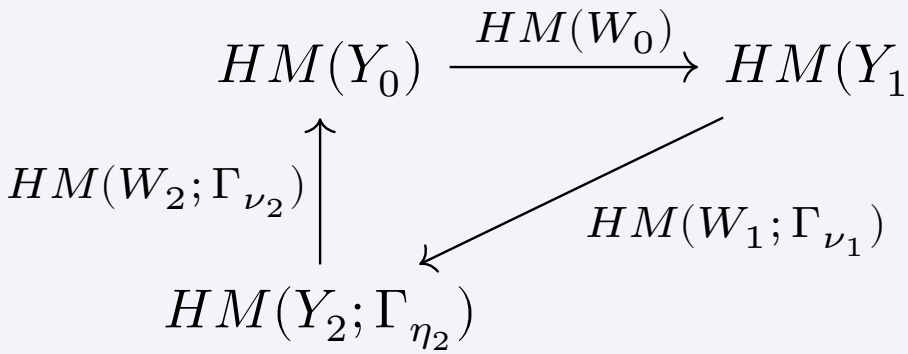
$$u^{(1+m)/2},\ u^{(1-m)/2}.$$

Their ratio is $u^m = -1$ at $u = -1$, so the pair cancels.

Main theorem

Theorem (Qiu–Y. '26)

Every surgery triad gives an exact triangle over $\mathbb{Z}$ at $u = -1$,



Here $\eta_2 \subset Y_2$ is the surgery core, ν_1 is the cocore disk of W_1, and ν_2 is the core disk of W_2. The statement holds for $\widehat{HM}$, $\widetilde{HM}$, $\overline{HM}$, and $\widetilde{HM}$.

Remark

After cyclic relabeling, $[\eta_2] = 0 \in H_1(Y_2; \mathbb{Z}/2)$. We also show

$$HM(Y_2;\Gamma_{\eta_2}) \cong HM(Y_2).$$

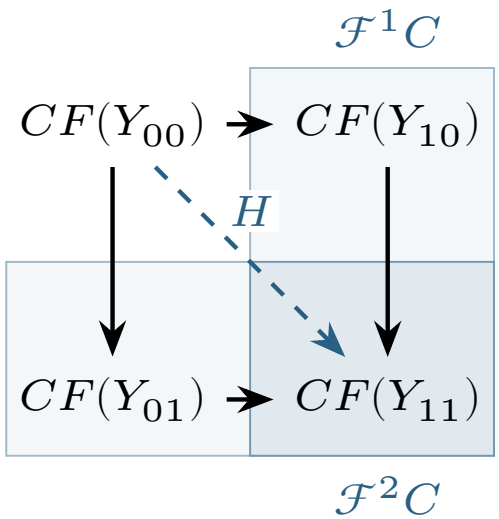
From triangles to spectral sequences

The surgery exact triangle follows from the chain level result

$$CF(Y_2) \simeq \mathrm{Cone}(CF(Y_0) \xrightarrow{f} CF(Y_1)),$$

with filtration $0 \subset CF(Y_1) \subset \mathrm{Cone}(f)$. For two crossings,

$$CF(Y_{22}) \simeq \mathrm{Cone}(CF(Y_{20}) \longrightarrow CF(Y_{21})) \simeq (C, D).$$



$$C = \bigoplus_{\epsilon', \epsilon'' \in \{0,1\}} CF(Y_{\epsilon'\epsilon''}),$$

$$\mathcal{F}^p C = \bigoplus_{\epsilon' + \epsilon'' \geq p} CF(Y_{\epsilon'\epsilon''}).$$

The homotopy H ensures $D^2 = 0$.

The cubical filtration gives

$$E_1 = \bigoplus_{\epsilon', \epsilon'' \in \{0,1\}} F(Y_{\epsilon'\epsilon''}) \Longrightarrow F(Y_{22}),\ d_1 = \sum_{\text{edges}} F(W).$$

Khovanov-to-Floer spectral sequences

For an n-crossing diagram, set $Y_v = -\Sigma_2(L_v)$, $v \in \{0,1\}^n$.

E_2	**Target**	**Coefficients**	**Authors**
$Kh^{\mathrm{red}}(L)$	$\widehat{HF}(-\Sigma_2(L))$	$\mathbb{F}_2$	Ozsváth–Szabó '05
$Kh^{\mathrm{red}}(L)$	$\widetilde{HM}(-\Sigma_2(L))$	$\mathbb{F}_2$	Bloom '11
$Kh^{\mathrm{red}}_{\mathrm{odd}}(L)$	$I^{\sharp}(-\Sigma_2(L))$	$\mathbb{Z}$	Scaduto '15

Scaduto uses surgery triangles with extra bundle data.

Conjecture (Ozsváth–Rasmussen–Szabó '13)

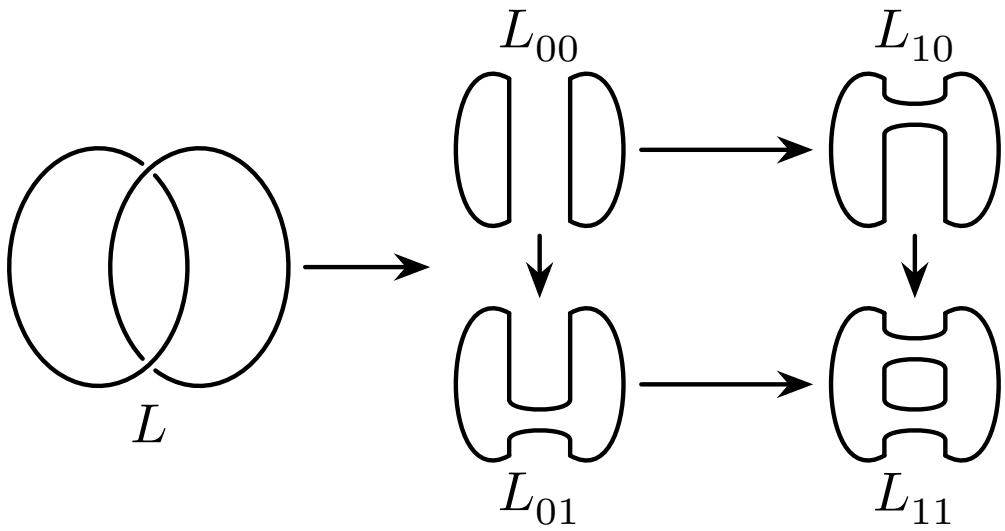
For any oriented link $L \subset S^3$, there is a spectral sequence

$$Kh^{\mathrm{red}}_{\mathrm{odd}}(L;\mathbb{Z}) \Longrightarrow \widehat{HF}(-\Sigma_2(L);\mathbb{Z}).$$

Odd Khovanov homology and unlinks

For the k-component unlink U_k, $\Sigma_2(U_k) \cong \#^{k-1}(S^1 \times S^2)$. Suppressing gradings,

$$Kh^{\mathrm{red}}(U_k;\mathbb{Z}) \cong Kh^{\mathrm{red}}_{\mathrm{odd}}(U_k;\mathbb{Z}) \cong \widehat{HF}(\Sigma_2(U_k);\mathbb{Z}) \cong \Lambda^*\mathbb{Z}^{k-1}.$$



On unreduced groups, a merging pair of pants identifies the two merging generators and uses multiplication in

$$\text{even } \mathbb{Z}[x_1,\dots,x_k]/(x_1^2,\dots,x_k^2),\ \text{odd } \Lambda^*\mathbb{Z}^k.$$

$$x_ix_j = x_jx_i,\ x_i \wedge x_j = -x_j \wedge x_i.$$

Even and odd Khovanov homology agree over $\mathbb{F}_2$.

Integral spectral sequence

Theorem (Qiu–Y. '26)

For an oriented link $L \subset S^3$, there is a spectral sequence over $\mathbb{Z}$, possibly depending on a chosen planar diagram of L,

$$E_2 = Kh^{\mathrm{red}}_{\mathrm{odd}}(L;\mathbb{Z}) \Longrightarrow \widetilde{HM}(-\Sigma_2(L);\mathbb{Z}).$$

The differentials d_r are odd in the mod 2 δ-grading.

Remark

Iteration gives local coefficients Γ_{ν_L} in the target, trivial at $u = -1$ since

$$[\nu_L] = 0 \in H_1(\Sigma_2(L);\mathbb{Z}/2).$$

Corollary (Qiu–Y. '26)

Kutluhan–Lee–Taubes '10–12 give $\widetilde{HM} \cong \widehat{HF}$ over $\mathbb{Z}$, proving the Ozsváth–Rasmussen–Szabó conjecture. The route $HM \cong ECH \cong HF$ via Taubes '08 and Colin–Ghiggini–Honda '11–12 is proved over $\mathbb{F}_2$.

Torsion and the choice of coefficients

Example (Shumakovitch '11)

Both $Kh^{\mathrm{red}}_{\mathrm{odd}}(9_{46};\mathbb{Z})$ and $Kh^{\mathrm{red}}_{\mathrm{odd}}(10_{140};\mathbb{Z})$ contain 3-torsion.

They are thin when 3 is invertible.

Example (Ozsváth–Rasmussen–Szabó '13)

$Kh^{\mathrm{red}}_{\mathrm{odd}}(9_{42};\mathbb{Q})$ is thin, while $Kh^{\mathrm{red}}_{\mathrm{odd}}(9_{42};\mathbb{F}_2)$ is not thin.

Thin means supported in one δ-grading. Over any commutative ring R, thin $Kh^{\mathrm{red}}_{\mathrm{odd}}(L;R)$ implies collapse at E_2.

For fixed Y, contact invariants in $\widehat{HF}(-Y;\mathbb{F}_2)$ have finitely many values. Over $\mathbb{Z}$, they distinguish infinitely many contact isotopy classes on T^3 (Massot '12).

Construction of local coefficients

Let X be a manifold, $f : X \to \mathbb{R}/\mathbb{Z}$ continuous, and $R = \mathbb{Z}[u, u^{-1}]$. Define $\Gamma_u = f^*\Gamma_{S^1}$ with formal rank-one fibers

$$\Gamma_u(x) = u^{f(x)}R.$$

Here Γ_{S^1} has monodromy u. Changing a real lift of $f(x)$ by an integer gives the same module. For a path γ, choose a real lift $\widetilde{f \circ \gamma}$. Transport is multiplication by

$$u^{\widetilde{f\circ\gamma}(1)-\widetilde{f\circ\gamma}(0)}.$$

Loop monodromy is $u^{\deg(f\circ\gamma)}$, giving $H_*(X; \Gamma_u)$.

Kronheimer–Mrowka '11, '17, '19 use this construction in instanton homology.

Our construction for monopole theory

On the blown-up configuration space $\mathcal{B}^\sigma(Y,\mathfrak{s})$, fix an integral 1-cycle $\eta \subset Y$ and a unit section $\tau_{\det}(\eta)$ of the determinant line. Define

$$f : \mathcal{B}^\sigma(Y,\mathfrak{s}) \to \mathbb{R}/\mathbb{Z},\ \ f(\mathbf{B}) = \frac{1}{2}\frac{i}{2\pi}\int_{\tau_{\det}(\eta)} B^t \pmod{\mathbb{Z}}.$$

The fiber is $u^{f(\mathbf{B})}\mathbb{Z}[u,u^{-1}]$. Gauge transformations change $\frac{i}{2\pi}\int B^t$ by an even integer, so f is well-defined.

Kronheimer–Mrowka '07 use $\mathbb{Z}[\mathbb{R}]$ and weights $t^{r_\eta(z)}$, where $r_\eta(z) = \frac{i}{2\pi}\int_{[0,1]\times\eta} F_{A_z^t}$. We use $u^{r_\eta(z)/2}$, formally corresponding to $u = t^2$.

Freeman '21 works over $\mathbb{Z}[\sqrt{-1}]$ with $t = \sqrt{-1}$. We use $u = -1$ over $\mathbb{Z}$. For a closed oriented surface $\nu \subset W$, the weight is

$$u^{(\langle c_1(\mathfrak{s}),[\nu]\rangle + \nu\cdot\nu + \chi(\nu))/2}.$$

The exponent is integral since $\chi(\nu)$ is even and

$$\langle c_1(\mathfrak{s}),[\nu]\rangle \equiv \nu\cdot\nu \pmod{2}.$$

Thanks for your attention.

Slides on floer.wiki/en